

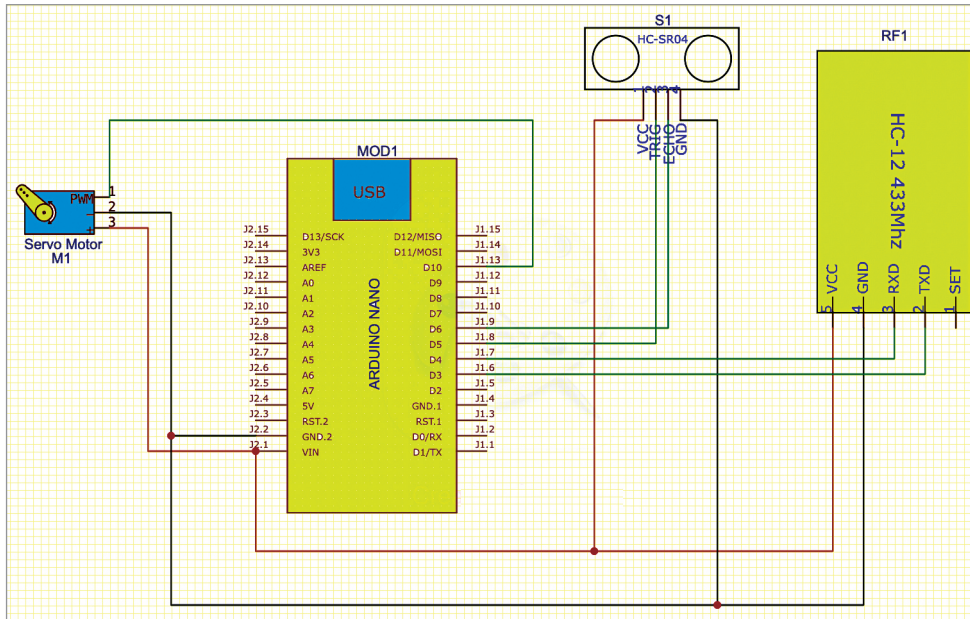


Fig. 3: Transmitter circuit

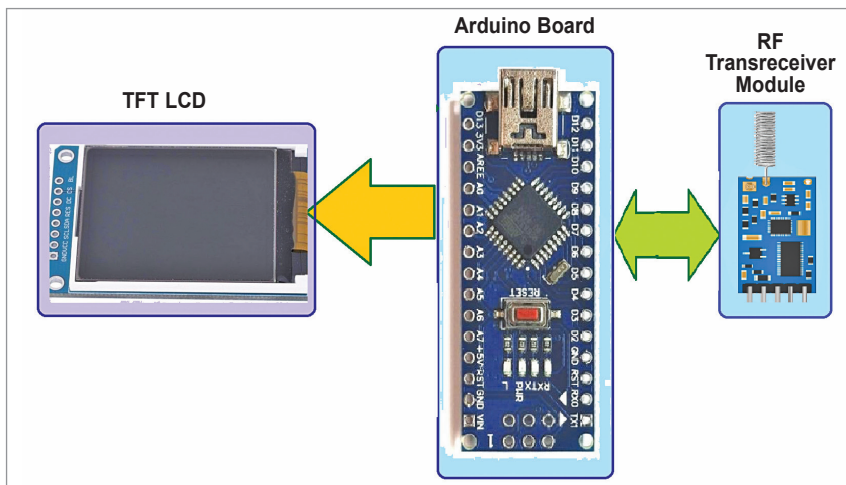


Fig. 4: Block diagram of receiver section

BILL OF MATERIALS

Components	Quantity
Arduino Nano/Uno (MOD1, MOD2)	2
Micro servo motor (M1)	1
Ultrasonic sensor HC-SR04 (S1)	1
RF transceiver HC-012 (RF1, RF2)	2
10.9cm (4.3-inch) TFT LCD (LCD1)	1
Resistors R1-R6 (1-kilo-ohm)	6
Resistors R7-R12 (2.2-kilo-ohm)	6
5V regulated power supply	2 (for transmitter and receiver)
Jumper wires and wires	As required

Radar transmitter section.

The transmitter section consists of an ultrasonic sensor (HC-SR04), a servo motor, an Arduino Nano, and an RF transceiver module. The block diagram of the transmitter section is shown in Fig. 2. A brief description of the components used is given here.

The ultrasonic sensor HC-SR04 is used to transmit ultrasonic sound and receive the reflected sound from any object in front of

it. It detects any object in its path and measures its distance. It provides a pulse width modulated (PWM) output, with the pulse width proportional to the object's distance. The servo motor rotates the ultrasonic sensor from 0-180° back and forth to continuously scan the semi-circle area. The RF transceiver module is used to serially transmit the distance and angle of the detected object, using a 434MHz carrier frequency.

The Arduino Nano is the main component of this section. It measures the width of the output pulse from the ultrasonic sensor and calculates the

object's distance. It rotates the servo motor from 0° to 180° and back by applying a PWM signal. It then serially transmits the object's distance and angle using the RF transceiver module.

Circuit diagram of radar transmitter section. Fig. 3 shows the circuit diagram of the transmitter. It consists mainly of four components: the HC-SR04 sensor, the servo motor, the HC-12 RF transceiver module, and the Arduino Nano board. Their interconnections are as follows:

The HC-SR04 sensor module has four interfacing pins: Vcc, Gnd, Trig, and Echo. The Vcc pin is connected to the 5V output pin of the Arduino board, and the Gnd pin is connected to the common circuit ground. The Trig and Echo pins are connected to digital pins D5 and D6 of the Arduino board, respectively.

The servo motor has three pins: Vcc, Gnd, and Signal. The Vcc pin is connected to the 5V output pin of the Arduino board, and the Gnd pin is connected to the common circuit ground. The Signal pin is connected to the PWM output pin D11 of the